Katherine Piper

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Education

Carnegie Mellon University | Pittsburgh, Pennsylvania — August 2024 – Present
Doctor of Philosophy | Computational Mechanics — Est. April 2029
Master of Science | Computational Mechanics — December 2025

California Institute of Technology | Pasadena, California — September 2019 – June 2024
Bachelor of Science | Mechanical Engineering

Research Experience

Carnegie Mellon University | Pittsburgh, Pennsylvania
Graduate Researcher — August 2024 – Present
- Constrained optimization machine learning methods for physics and engineering problems

DEVCOM Army Research Laboratory | Aberdeen, Maryland
Research Associate (Advisors: Jarek Knap and Kevin LoGrande) — Summer 2026
- Advancing Diffusion Maps Surrogates with Constrained Optimization

NASA Jet Propulsion Laboratory | Pasadena, California
Academic Part Time Employee — October 2023 – August 2024
Year Round Intern — May 2023 – September 2023
- Additive Manufacturing Center (AMC) and Two-Phase Thermal Technology Laboratory
- Methods led to 100 times cost reduction and 42% performance improvement over state-of-the-art TPL design
- Two NASA New Technology Reports and US Patent submitted

California Institute of Technology | Pasadena, California
Summer Undergraduate Research Fellowship — June 2022 – June 2024
Research Advisor – Vinamra Agrawal, Los Alamos National Laboratory
- Implementation of a model for ductile fracture in functionally graded additively manufactured metals

Awards

CEE Research Spotlight: 1 of 4 Featured Posters — August 2026
CMU CEE Fall 2026 Poster Competition

Journeyman Fellowship — June-August 2026
DEVCOM Army Research Laboratory

Grand Prize: Sustainable Impact Award – $4,000 — March 2026
Gresham Smith Innovation Hackathon

Graduate Fellowship — 2024 - 2029
Carnegie Mellon University College of Engineering

Research Grants

ACCESS – EXPLORE – National Science Foundation Computing Grant — September 2024 – Present

Publications and Patents

(Under Review) K. Piper and S. N. Roberts. Beyond Porosity and Permeability: An Alternative Design Space for Passive Capillary Evaporators. September 2026. Preprint available at https://doi.org/10.1184/R1/34814499

(Provisional Patent) K. Piper, S. N. Roberts, and T. Daimaru. Novel Geometric Considerations for Solid and Porous Design in Loop Heat Pipe Evaporators. September 2025. *Assigned to California Institute of Technology*

(In Preparation) K. Piper, L. T. Biegler, N. Trask, and K. Dayal. Direct Nonlinear Optimization Enables Reduced Order Models for Contact.

K. Piper and V. Agrawal. Ductile fracture in functionally graded materials: Insight into crack behavior within the gradient interface. arXiv preprint arXiv:2411.18642 (2024)

Media Coverage and Public Engagement

Students pitch sustainability innovations at inaugural Hackathon — March 4, 2026
Carnegie Mellon University Press Release - Project Featured as First Place Team

Loop Heat Pipes for Smallsat Swarms (LHPss) — October 2024
Scott Roberts: NASA JPL FY24 Progress Report - Named Key Contributor